Michele Myong, PhD

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SUMMARY

I am a materials and physical chemist with 9+ years of R&D experience in optical materials, ultrafast laser spectroscopy, and accelerator science. Currently, I work in industry as a systems integration engineer developing products that use enhanced optical spectroscopy for high-resolution detection of target molecules. I also develop deep learning architectures, using machine learning to improve the analysis of spectroscopy data. Passionate about leveraging my skills in laser science and engineering, I seek a research role at a national laboratory that draws on my experience in both industry and national labs.

SKILLS

Spectroscopy: electron accelerator and ultrafast laser-based science, time-resolved fluorescence, transient absorption microscopy, kinetic modeling, (TD-)DFT calculations, infrared and vibrational spectroscopy

Materials Characterization/Chemistry: X-ray diffraction, atomic layer deposition, electron microscopy, electrochemistry, thermogravimetric analysis, dynamic light scattering, film fabrication (spin coating), organic synthesis
Engineering: test engineering, system integration, experiment validation, data management infrastructure
Languages and Platforms: Git, SQL, MATLAB, Python, PyTorch, Igor, JAX, Linux, TensorFlow, DeepChem
Python Libraries: numpy, pandas, scikit-learn, scipy, matplotlib, seaborn
Machine Learning: Linear/Logistic Regression, Bayesian Statistics, Principal Component Analysis, Neural Network
OS and Platforms: GitHub, Linux, Jupyter Notebook, Marimo, WSL, Visual Studio Code, Google Colab
Soft Skills: coordinating efforts across multidisciplinary teams, effective technical presentations, proposing and leading new experiments, developing concepts and proposals, mentoring scientists and technical staff

SELECTED PROJECTS

REDOX LADDER OF ENERGY MATERIALS (Brookhaven National Lab)

- Created a new measurement method to measure conjugated polymers (P3HT) for energy applications using accelerator science in a multi-user experimental facility (Laser Electron Accelerator Facility)
- Benchmarked redox levels of state-of-the-art optoelectronic materials (PM6, Y6) for improved materials design
- Analyzed complex kinetics data to determine exact experimental parameters that allow charge transfer between radical ions for highly precise energy measurements

DEVELOPING A NEW PLATFORM FOR QUANTUM SENSING (Brookhaven National Lab)

- Established pulse radiolysis as a technique to study room temperature qubits (spin-correlated radical pairs) for quantum information science (QIS)
- Designed and developed a magnetic-field dependent emission experimental setup to measure quantum beats
- Observed consistent evidence of quantum beats and extended this to other molecular systems for QIS

PROFESSIONAL EXPERIENCE

PICARRO, INC. Santa Clara, CA
Systems Integration Engineer Feb 2026 – Present

- Constructed benchtop prototype of a mobile gas monitoring application to validate the peak detection algorithm for high-resolution, automated detection of gases in real-time; coordinated efforts between software, firmware, electrical, and mechanical engineers
- Designed the hardware setup for a large-scale testbed experiment to measure the wind trajectories of trace amounts of gases to improve technology that pinpoints gas leaks in real-time
- Built deep learning transformer models for one-shot IR-to-molecule prediction to improve the experimental workflow that produces cavity-ringdown spectroscopy (CRDS) spectra of target molecules for the company-wide spectral library
- Validated the use of the Kalman filter algorithm for improved analysis on company wide data and developed a new metric to quantify its improvements over previous methods

BROOKHAVEN NATIONAL LABORATORY Long Island, NY
Postdoctoral Research Associate Sep 2021 – Sep 2025
Joanna Fowler Award in the Chemical Sciences

- Developed a new measurement method at the Laser Electron Accelerator Facility (LEAF) that uses pulse radiolysis and accelerator science to measure sustainable energy materials for to millivolt precision in low-polarity environments

- Constructed magnetic-field dependent experiment at the electron accelerator for molecular quantum information science, making the first observation of quantum beats using electron pulses for room temperature qubits
- Built a database of radical ion spectra and created a data workflow to integrate new experimental spectra into the existing data structure; trained a neural network to predict molecule candidates from optical absorption spectra
- Elucidated a complex photocatalysis mechanism in collaboration with BioLEC EFRC (Princeton), established polarons as dynamic probes of nanoscale environments, and interpreted data from electron transfer experiments in the context of new physical models, resulting in multiple published papers

LEADERSHIP EXPERIENCE

CHEMISTRY POSTDOC SEMINAR SERIES — Long Island, NY
Creator and Organizer — Sep 2021 – Mar 2023

- Founded the first ever chemistry postdoc seminar series at Brookhaven National Laboratory
- Organized and advertised 30+ seminar presentations and a grad school application panel for SULI summer interns

EDUCATION

NORTHWESTERN UNIVERSITY — Evanston, IL
Doctor of Philosophy | Advisor: Prof. Michael Wasielewski — Aug 2016 - Aug 2021
Physical & Materials Chemistry
NSF Graduate Research Fellow

COLUMBIA UNIVERSITY — New York, NY
Bachelor of Arts in Chemistry — Aug 2012 - May 2016
Société de Chimie Industrielle Fellow
Relevant Coursework: Materials Chemistry, Quantum Mechanics, Data Structures and Algorithms

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

DEEP LEARNING CERTIFICATE — May 2025
Erdős Institute project link; predicted drug efficacy of small molecules using graph neural networks

DATA SCIENCE CERTIFICATE — December 2024
Erdős Institute project link; automated analysis of NASA chemical data with machine learning

RESEARCH SOFTWARE ENGINEERING (INTERSECT PROGRAM) — Princeton, NJ
Princeton University; implementing best software practices — July 2025
One of 30 participants selected and fully funded by NSF grant

SELECTED PUBLICATIONS

1. **Myong, M. S.**; Bird, M. J.; Miller, J. R. "The Kinetics and Energetics of Electron Transfer to Dimer Radical Cations." *J. Phys. Chem. B*, 2023, 127, 13, 2881-2886
2. **Myong, M. S.**; Qi, Y.; Stern, C. L.; Wasielewski, M. R. "Ultrafast Photo-Driven Charge Transfer Exciton Dynamics in Mixed-Stack Pyrene-Perylenediimide Single Cocrystals." *J. Mater. Chem. C* 2021, 9, 16911-16917
3. **Myong, M. S.**; Young, R. M.; Wasielewski, M. R. "Excimer Diffusivity in 9,10-Bis(phenylethynyl) anthracene Assemblies on Anodic Aluminum Oxide Membranes." *J. Phys. Chem. C* 2021, 125, 44, 24498-24504
4. **Myong, M. S.**; Zhou, J.; Young, R. M.; Wasielewski, M. R. "Charge-Transfer Character in Excimers of Perylenediimides Self-Assembled on Anodic Aluminum Oxide Membranes." *J. Phys. Chem. C* 2020, 124, 8, 4369–4377

SELECTED PRESENTATIONS

1. **Myong, M. S.**; Bird, M. J. "Referenced Redox Potentials of Hole Acceptors in Nonpolar Media" Materials Research Society Spring National Meeting, Seattle, WA, April 2024, oral presentation
2. **Myong, M. S.**; Bird, M. J. "Fundamental Chemical Insights on Molecules and Materials for Sustainable Energy: From Photo- to Electron-Induced Processes." Texas A&M University, Department of Chemistry, December 2023, invited talk.
3. **Myong, M. S.**; Bird, M. J. "Fundamental Chemical Insights on Molecules and Materials for Sustainable Energy: From Photo- to Electron-Induced Processes." University of Texas at Arlington, Department of Chemistry, January 2023, invited talk.
4. **Myong, M. S.**; Bird, M. J. "Fundamental Chemical Insights on Molecules and Materials for Sustainable Energy: From Photo- to Electron-Induced Processes." Baylor University, Department of Chemistry, December 2022, invited talk